Piyush Maji

MSc Physics

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Degree	University/Institute	Year	CGPA/Marks(%)
MSc Physics	IIT Hyderabad	2027	7.5
BSc Physics	Burdwan University	2024	7.28
XII (West Bengal Council of Higher Secondary Education)	Sugandha High School	2021	75.00%
X (West Bengal Board of Secondary Education)	Jadabpur High School	2019	86.00%

SKILLS

Programming, Data & Frontend: Python, SQL, NumPy, Pandas, SQLite, Bash, HTML, CSS, JavaScript
Software & Deployment: FastAPI, Flask, Docker, AWS EC2, Git, GitHub Actions, pytest
ML & DL: scikit-learn, LightGBM, XGBoost, PyTorch, PyTorch Geometric, CNNs, RNNs, Transformers, GNNs, TCNs
Model Evaluation & Explainability: SHAP, GNNExplainer, Statistical Metrics, Time-Series Validation
GenAI & Retrieval: LangGraph, RAG, ChromaDB, Gemini API

PROJECTS

FraudNet | AI-Powered Fraud Investigation Completed
Python | PyTorch Geometric | LangGraph | ChromaDB | FastAPI | AWS
- Compared GNN and tabular models across three time windows; Random Forest averaged 0.862 F1 and 0.890 PR-AUC.
- Built a LangGraph workflow combining GNNExplainer, graph analysis, ChromaDB RAG, and LLM-generated reports.
- Deployed Dockerized FastAPI on AWS EC2 with an investigation dashboard and automated tests.

DemandCast | Retail Demand Forecasting Completed
Python | PyTorch | Attention GRU | LightGBM | SARIMA | Prophet | MySQL
- Built an attention GRU forecasting 14-day sales from 56-day histories across 1,115 retail stores.
- Reduced WMAPE from 20.92% to 20.13% versus a matched seasonal baseline in a single-origin backtest.
- Integrated SQL forecast storage and an interactive dashboard with store filters, data checks, and report exports.

DigitalTwinAI | Hydraulic Condition Monitoring Ongoing
Python | PyTorch | Temporal Convolutional Networks | Flask | JavaScript
- Built a multirate TCN using 14 sensor channels and 140 summary features from 2,205 measured hydraulic cycles.
- Achieved 0.9449 mean component macro-F1 across three grouped folds for hydraulic condition classification.
- Built a Flask/JavaScript dashboard for sensor replay, condition estimates, validation plots, and CSV exports.

Spam Classification from Scratch Completed
Python | NumPy | Pandas | MySQL | Matplotlib
- Implemented Naive Bayes and Logistic Regression from scratch in NumPy for 5,572 labeled SMS messages.
- Built BoW/TF-IDF features, Laplace smoothing, log-space inference, and L2 gradient descent with early stopping.
- Recorded 92.86% Naive Bayes F1 and 0.9811 Logistic Regression ROC-AUC on 1,115 evaluation messages.

EXPERIENCE

Software Quality Assurance Shifter | Belle II Collaboration | KEK, Japan Ongoing
- Reviewed nightly simulation and reconstruction plots against reference distributions to identify software regressions.
- Maintained validation records, reported discrepancies to developers, and tracked unresolved issues through follow-up.

Data Production Shifter | Belle II Collaboration | KEK, Japan Ongoing
- Monitored distributed computing and data-production workflows, recording operational data, faults, and recurring issues.
- Analyzed monitoring errors and presented findings in weekly team meetings to support troubleshooting and resolution.

RESEARCH

Bayesian Optimization for Accelerator Parameter Tuning | Belle II Collaboration, KEK, Japan Ongoing
Prof. Sourav Sandilya | IIT Hyderabad
- Built a Gaussian Process optimizer with EI/UCB acquisition, parameter constraints, and proximal biasing.
- Designed the frontend; validated functionality through combined-function tests and real-accelerator testing.
- Raised median synthetic efficiency from 92.19% to 99.95% in 40-trial Hartmann4D tests by tuning a fourth parameter.

Graph-Based Full Event Interpretation (graFEI) | Belle II Collaboration, KEK, Japan Feb–Aug 2026
Prof. Sourav Sandilya | IIT Hyderabad
- Built particle-event graphs in PyTorch Geometric for GNN-based B-meson decay reconstruction.
- Studied joint node classification for particle identity and edge classification for ancestor-stage (LCAS) prediction.
- Prepared basf2 Monte Carlo inputs and evaluation workflows for particle identification and decay-tree accuracy.

POSITIONS OF RESPONSIBILITY

Placement Coordinator | Office of Career Services, IIT Hyderabad May 2026–Present
- Engaged recruiters and coordinated campus placements, Japan Career Day 2026, and Career Fair 2026.

Inbound & Logistics | International Relations Cell, IIT Hyderabad Aug 2025–Present
- Coordinated visits from eight Japanese universities and facilitated interactions with DNP and Mitsubishi delegates.

Coordinator | Extra Mural Lectures, IIT Hyderabad May 2026–Present
- Managed ₹100K+ budgets and event logistics; hosted astronaut Shubhanshu Shukla and Ericsson's CTO.

Event Coordinator & Host | AI4SE, IIT Hyderabad July 2026
- Coordinated and hosted AI4SE 2026, bringing together researchers, educators, and industry professionals.

RELEVANT COURSES

ML for Physical Sciences • Machine Learning for HEP •Computational Physics • Linear Algebra • Statistical Mechanics